CASE STUDY 4

Unit IV: Hashing

Dictionary Application Using a Hash Table

Aim

To store and retrieve words and meanings using hashing with collision resolution.

Objectives

- Understand hash tables and hash functions.
- Store word-meaning pairs.
- Handle collisions using chaining.

Theory

A hash table stores data at an index calculated by a hash function. A dictionary can store a word as a key and its meaning as a value. Different words may map to the same index; this is called a collision. In chaining, each table position stores a list of entries, allowing multiple keys at the same index.

Algorithm

1. Create a table containing empty lists.
2. Calculate an index from each word using a hash function.
3. Store the word and meaning in the list at that index.
4. To search, calculate the index and check the entries in that list.

Python Program

```
# Dictionary using a hash table with chaining
class HashDictionary:
    def __init__(self, size=10):
        self.size = size
        self.table = [[] for i in range(size)]

    def hash_function(self, word):
        return sum(ord(ch) for ch in word) % self.size

    def insert(self, word, meaning):
        index = self.hash_function(word)
        for item in self.table[index]:
            if item[0] == word:
                item[1] = meaning
                return
        self.table[index].append([word, meaning])

    def search(self, word):
        index = self.hash_function(word)
        for item in self.table[index]:
            if item[0] == word:
                return item[1]
        return "Word not found"

dictionary = HashDictionary()
dictionary.insert("algorithm", "Steps to solve a problem")
dictionary.insert("data", "Facts or information")
dictionary.insert("structure", "A way to organize data")

print("Meaning:", dictionary.search("algorithm"))
print("Meaning:", dictionary.search("data"))
print("Meaning:", dictionary.search("computer"))
```

Sample Output

```
Meaning: Steps to solve a problem\nMeaning: Facts or information\nMeaning: Word not found
```

Conclusion

A hash table can retrieve dictionary entries quickly on average. Chaining helps handle collisions when different words map to the same index.